

Emergence of human pathogenic fungi in a changing climate

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As climate variability intensifies, human-pathogenic fungi are spreading into new regions and causing more disease. Addressing emergence will require stronger surveillance and forecasting; research on fungal dispersal, adaptation and population immunity; and policies that support vaccine development, antifungal stewardship and diagnostic access.

Human fungal infections are rapidly rising within and beyond historically endemic regions. Notable examples include rising *Coccidioides* infections in the Southwestern USA, the spread of *Cryptococcus gattii* across the Pacific Northwest, and the largest reported *Blastomyces* outbreak in 2023. Because these fungi are maintained in environmental reservoirs, their distributions and transmission are linked to environmental conditions (Fig. 1a). Climate change is reshaping where pathogenic fungi persist and how frequently they cause illness. Rising temperatures and more frequent wet–dry cycles enhance fungal growth and facilitate dispersal of infectious spores (Fig. 1b). Aridification contributes to dust-related *Coccidioides* exposure¹ and flooding can lead to invasive mold infections². Warming may favour heat-tolerant fungi, potentially eroding the human body's thermal barrier against infection³. Our ability to predict, prepare for and mitigate climate-driven emergence of fungal pathogens remains constrained by limitations in data availability, scientific understanding and public health preparedness. Here, we outline priorities for addressing these constraints (Fig. 1c).

Enhancing surveillance

Lack of longitudinal and spatially comprehensive data on fungal diseases hinder our ability to document emergence and assess climate impacts. Priorities include stronger surveillance of human cases, the environment and sentinel animals.

Few countries conduct human surveillance for fungal diseases³, and reporting remains limited. In the USA, fewer than half of states report coccidioidomycosis and only eight report blastomycosis. Existing data likely undercount coccidioidomycosis cases, particularly among high-risk groups such as outdoor workers, who face elevated exposures and limited healthcare access. Diagnostics are scarce in low- and middle-income countries, with *Aspergillus* antibody and *Histoplasma* antigen tests being absent from nearly all African countries despite high disease burdens^{3,4}.

Longitudinal environmental surveillance can help identify climate-driven fungal range expansion and exposure hotspots before clinical cases accumulate. Wastewater and vector surveillance have revealed geographic expansion for poliovirus and mosquito-borne

diseases, respectively, but fungal monitoring remains limited, because fungi are difficult to detect in the environment. Emerging approaches hold promise: wastewater surveillance has detected *Candida auris* in hospital sewersheds; air-sampler networks have characterized spatial variation in airborne *Coccidioides* across Phoenix, Arizona; and citizen science surveys of soil samples have enabled broad-scale surveillance of triazole-resistant *Aspergillus fumigatus* across the UK.

Finally, since all major human fungal pathogens also infect animals, monitoring companion animals, livestock and wildlife can reveal disease patterns and emergence in regions with limited human reporting. Veterinary cases of *C. gattii* preceded human ones in British Columbia and canine serosurveillance for coccidioidomycosis has revealed disease hotspots in US states lacking human reporting.

Research needs

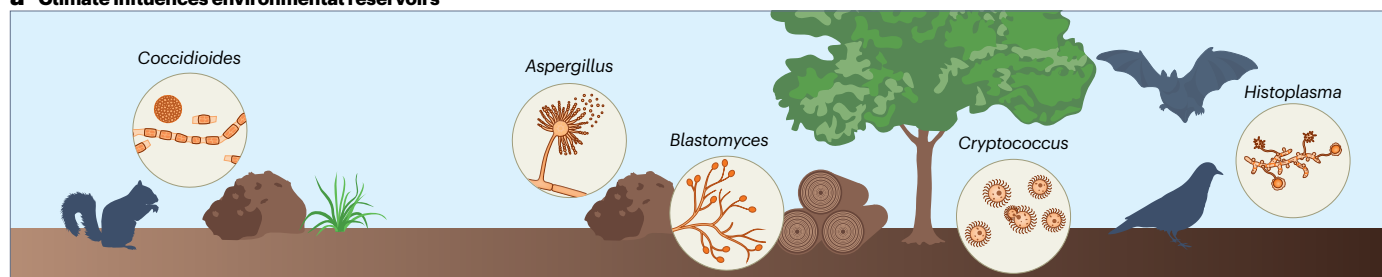
Beyond data needs, several knowledge gaps constrain our ability to understand and predict emergence. Research priorities should focus on understanding climate epidemiology, fungal adaptation and dispersal, and size of susceptible populations.

Although climate variability shapes fungal disease distributions, the climate epidemiology of many fungal pathogens remains poorly understood. Epidemiological and laboratory research is needed to establish how specific climate conditions drive fungal growth, dispersal and human exposure. For coccidioidomycosis, associations between wet–dry cycling and incidence are well characterized⁵, but for most other mycoses, mechanistic links between climate and transmission remain incompletely understood. Controlled laboratory experiments that isolate climate–response relationships are sparse. Robust characterization of climate–incidence associations would help determine how much of the observed increase in fungal disease incidence is driven by anthropogenic climate change.

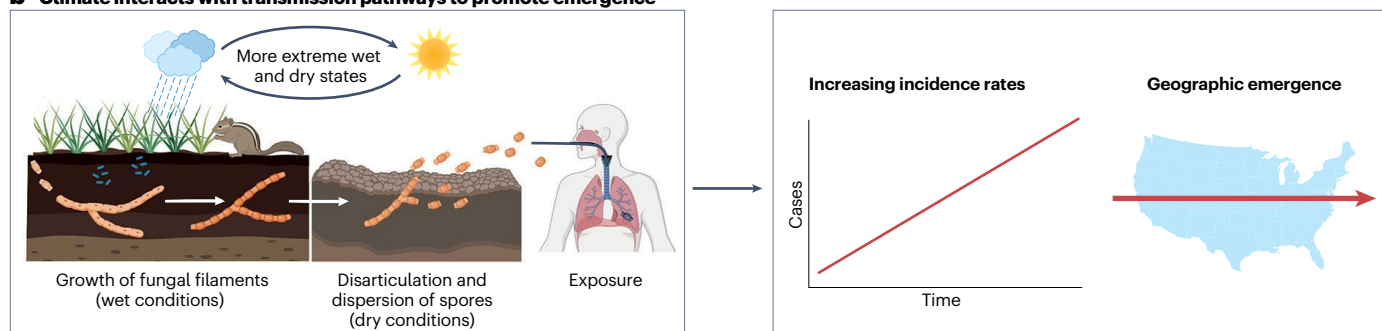
Even with well-characterized climate–incidence relationships, fungal adaptation to environmental stressors would challenge our ability to use historical patterns to predict future suitability ranges. For example, warming is expected to reduce some mosquito-borne diseases where temperatures exceed mosquito thermal limits, but rapid mosquito adaptation may prevent these declines. Fungal thermal adaptation remains underexplored, but heat stress-induced genomic mutations in *Cryptococcus deneoformans* have been documented⁶.

Understanding fungal spore dispersal is key to predicting climate-driven shifts in fungal pathogens, yet the relative contribution of wind- and animal-mediated spread remains unclear for many fungal species. For *Coccidioides*, rodents are debated as primary reservoirs that may carry pathogens to new locations, although increased infections following major wind and dust events also support long-distance, wind-driven dispersal. Bat and bird guano are associated with *Histoplasma* and *Cryptococcus* spp., but their role in transporting fungal

a Climate influences environmental reservoirs



b Climate interacts with transmission pathways to promote emergence



c Understanding and responding to emergence

Data	Research questions	Public health action
Strengthened global surveillance and diagnostics in humans, animals and the environment 	- What is the relationship between climate and specific fungal diseases? - How much of the increases in diagnoses are attributable to anthropogenic climate change? - How will fungal adaptation alter expected patterns of emergence? - How will fungi disperse to locations of emerging suitability? - What is the size of the susceptible population in endemic and emerging areas?	- Vaccine development - Laws and education aimed at faster diagnoses and reduced antifungal resistance - Short-term forecasts and long-term projections 

Fig. 1 | **a**, Several human-pathogenic fungi live in the environment and may cause disease in otherwise healthy individuals. **b**, Example of how intensification of climate change-driven wet–dry cycles may increase emergence

of coccidioidomycosis (Valley fever). **c**, Priorities for data, research and public health action. Left panel of part **b** reprinted with permission from ref. 5, Elsevier.

pathogens remains unclear. Genomic and phylogeographic analyses may reveal landscape features or pathways that facilitate long-distance spore dispersal. Genomic analyses suggest that international tulip bulb shipping contributed to geographic emergence of azole-resistant *A. fumigatus*⁷.

Finally, limited data of population immunity to mycoses prevents the understanding of the susceptible population size under climate-driven fungal expansion. Existing estimates of population immunity against coccidioidomycosis and histoplasmosis in the USA are often decades old. Documenting natural immunity is challenging because measures of cellular immunity (for example, skin tests) have low sensitivity and humoral immunity against many fungi is not retained.

Public health priorities

Amidst ongoing and anticipated emergence, the public health community must be prepared to protect growing at-risk populations. We identify four strategies.

First, no licensed human vaccines exist for any fungal pathogen, making vaccine development a key priority. Recent candidates have shown promise: a *Candida albicans* vaccine in phase I clinical trial and a *Coccidioides* live attenuated vaccine in dogs. Second, rapid rises in antifungal resistance have accompanied fungal disease emergence. Extensive antifungal use in agriculture has been linked to azole-resistant *A. fumigatus*⁸. Legislation restricting antifungal use in agriculture and

developing separate antifungals for clinical and agricultural use is warranted.

Third, early warning systems, analogous to forecasting models for malaria, West Nile virus and dengue, are needed to predict fungal disease risk and guide public health intervention. As a proof of concept, a short-term, climate-based forecast for coccidioidomycosis enabled alerts for increased clinical vigilance during a high-transmission season⁹. Longer-term projections estimate where and how severely fungal disease burden will increase under different emissions scenarios, motivating development of diagnostic capacity, clinical training and community awareness before emergence. For coccidioidomycosis, models predict expansion of endemic regions across North America by the end of the century¹⁰, and a lengthened transmission season in California. Projection studies for other fungal infections are needed.

Lastly, it is imperative to reduce diagnostic delays. Even in known endemic areas, fungal pneumonias are routinely misdiagnosed, causing diagnostic delays that can last months. Physicians in emerging areas are even less likely to consider fungal aetiology in patients without travel history. Physician education and legislation are needed in regions where cases are beginning to, or projected to, appear. California's Valley Fever Screening and Prevention Act of 2025 offers a useful model, mandating annual identification of high-incidence coccidioidomycosis regions and free screening tests for adults. Because populations most exposed to emerging fungal pathogens frequently have the least access to healthcare, equity must be central to interventions.

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Competing interests

The authors declare no competing interests.